Daniel Barta

Machine Learning Researcher

PERSONAL PROFIL

Data-driven engineer with a solid background in computer science and software engineering, experienced in machine learning, optimization, and data analytics.

CONTACT

+49 163 6081304
Berlin, Germany

daniel.barta@web.de

LinkedIn

TECHNOLOGIES

Languages:
Python, Java, MATLAB, Swift, JavaScript, C++

ML/AI Frameworks:
PyTorch, TensorFlow

Quantum Computing:
Qiskit

Tools/DevOps:
Docker, Git & GitHub, Linux/UNIX, SQL

LANGUAGES

German (Native)
English (Fluent)

SOFT SKILLS

Analytical Thinking
Problem-Solving
Clear Communication
Collaboration
Adaptability
Interdisciplinary Teamwork

PROFESSIONAL EXPERIENCE

Working Student – Quantum Computing & AI
Fraunhofer FOKUS, Berlin | Mar 2023 – Aug 2025

- Built diffusion-based pipeline for generation of parameterized quantum circuits
- Accelerated quantum benchmark runs with surrogate models
- Built neuro-symbolic reasoning modules for legally compliant autonomous driving
- Development of Backend and Frontend Services for Knowledge Graph
- Developed a Federated Search Service to query multiple databases

Intern – Engine and Data Analytics
BMW Group, Munich | Mar 2020 – Aug 2020

- Enhanced automated model verification pipelines, reducing testing time and improving accuracy
- Optimized database schemas for faster retrieval, improving analytics efficiency
- Analyzed driving-behavior datasets using Python and MATLAB with advanced game-theoretic models

EDUCATION

M.Sc. in Electrical and Information Engineering
Technical University of Berlin | Sep 2022 – Jan 2026

- *Thesis:* "Leveraging Diffusion Models for Parameterized Quantum Circuit Generation"
- *Coursework:* Quantum Computing, Machine Learning, Optimization, Information Theory and Applications, Foundations of Stochastic Processes, Foundations of Statistical Inference, Detection and Estimation

B.Sc. in Electrical and Information Engineering
Technical University of Munich | Oct 2018 – Mar 2022

- *Thesis:* "One-Bit Channel Estimation with Generative Adversarial Networks"
- *Coursework:* Signal Processing, Electrical Power Engineering, Stochastic Signals, Computational Intelligence

PUBLICATIONS

- **"Leveraging Diffusion Models for Parameterized Quantum Circuit Generation"**, presented at **IEEE QCE '25**. arXiv:2505.20863
- **"Surrogate Benchmark for Quantum Architecture Search"**, arXiv:2506.06762 , GitHub Repo
- **"Legally-Guided Automated Decision-Making System Using Language Model Agents for Autonomous Driving"**, in *Lecture Notes in Computer Science* (Springer), 2025. DOI:10.1007/978-3-031-72407-7_17